ELEC60008
EE9-CS1-1

IMPERIAL COLLEGE LONDON

DEPARTMENT OF ELECTRICAL AND ELECTRONIC ENGINEERING
EXAMINATIONS 2024

EEE/EIE PART III/IV: MEng, BEng and ACGI

CONTROL ENGINEERING

Monday, 9 December 10:00 am

Time allowed: 3:00 hours

There are THREE questions on this paper.

Answer ALL questions.

Question ONE, TWO, and THREE carry 40, 30 and 30 marks respectively.

Any special instructions for invigilators and information for candidates are on page 1.

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Second Marker(s) :

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CONTROL ENGINEERING

Information for candidates:

- The solution of $\dot{x}(t) = Ax(t) + Bu(t)$ is

$$x(t) = e^{A(t-t_0)}x(t_0) + \int_{t_0}^{t} e^{A(t-\tau)}Bu(\tau)d\tau.$$

- Given a matrix A, the matrix exponential is defined as

$$e^{At} = I + At + \frac{(At)^2}{2!} + \frac{(At)^3}{3!} + \dots$$

- The solution of $x(k+1) = Ax(k) + Bu(k)$ is

$$x(k) = A^{(k-k_0)}x(k_0) + \sum_{j=k_0}^{k-1} A^{k-j-1}Bu(j).$$

- The reachability matrix is $\mathscr{R} = [B\ AB\ \dots\ A^{n-1}B]$. A system is reachable if and only if $\text{rank}[\mathscr{R}] = n$. A discrete-time system is controllable if and only if $\text{rank}[A^n\ \mathscr{R}] = \text{rank}[\mathscr{R}]$.
- A system is reachable if and only if $\text{rank}\begin{bmatrix} A-\lambda I & B \end{bmatrix} = n$ for all $\lambda \in \sigma(A)$.
- The observability matrix is $\mathscr{O} = \begin{bmatrix} C \\ CA \\ \vdots \\ CA^{n-1} \end{bmatrix}$. A system is observable if and only if $\text{rank}[\mathscr{O}] = n$.
  A discrete-time system is reconstructable if and only if $\text{rank}\begin{bmatrix} A^n \\ \mathscr{O} \end{bmatrix} = \text{rank}[\mathscr{O}]$.
- A system is observable if and only if $\text{rank}\begin{bmatrix} A-\lambda I \\ C \end{bmatrix} = n$ for all $\lambda \in \sigma(A)$.
- Ackermann's formula: assume $m = 1$ (single-input system) and let $p_{des}(\lambda)$ be a desired closed-loop polynomial. Then
$$K = -[0\ 0\ \dots\ 1]\mathscr{R}^{-1}p_{des}(A)$$
where $\mathscr{R}$ is the reachability matrix, is such that characteristic polynomial of $A + BK$ is $p_{des}(\lambda)$.

1. Consider the following short questions accounting, overall, for 40% of the exam mark (that is 40 marks out of 100).

a) Consider the discrete-time linear system described by the equation

$$x(k+1) = \begin{bmatrix} -5 & -9 \\ 4 & 7 \end{bmatrix} x(k).$$

Assess the stability of the equilibrium point of the system. [5 marks]

b) Consider the continuous-time linear mass-damper system described by the equation

$$\dot{x} = \begin{bmatrix} 0 & 1 \\ -\frac{K}{M} & -\frac{F}{M} \end{bmatrix} x + \begin{bmatrix} 0 \\ \frac{1}{M} \end{bmatrix} u.$$

Let $M > 0$ and $F > 0$. Study the equilibrium points of the system as a function of K and $u(t) = \bar{u}$ for all t, with $\bar{u} \neq 0$. [5 marks]

c) Consider the system at point (b) with $K \geq 0$, $F \geq 0$ and $M > 0$. Explain in each of the following cases, if the zero equilibrium point is asymptotically stable, stable, or unstable:

i) $F = 0, K > 0$.

ii) $F < 2\sqrt{KM}, K > 0$.

iii) $F = 2\sqrt{KM}, K > 0$.

iv) $F > 2\sqrt{KM}, K > 0$.

v) $F > 0, K = 0$.

vi) $F = 0, K = 0$.

[6 marks]

d) Consider the continuous-time nonlinear system described by the equation

$$\dot{x}_1 = \alpha x_1^3 - x_2$$
$$\dot{x}_2 = \alpha x_2^3 + x_1$$

where α is a real number.

i) Compute all the equilibrium points of the system. [2 mark]

ii) Assess the stability properties of the zero equilibrium point of the nonlinear system using the linearization as a function of α. [4 marks]

iii) Consider the function $V(x) = x_1^2 + x_2^2$ representing a circle in the phase portrait. Compute the derivative of V with respect to time, namely $\dot{V} = \frac{\partial V(x)}{\partial x}\frac{dx}{dt}$ and determine the stability properties of the zero equilibrium point of the nonlinear system as a function of α. [4 marks]

e) Consider a linear, time-varying, discrete-time system described by the equation

$$x(k+1) = A(k)x(k).$$

A common belief is the following.

(C) If the matrix $A(k)$ has constant eigenvalues with absolute value smaller than one then the linear, time-varying, system is asymptotically stable.

To disprove the claim (C) consider the matrix

$$A(k) = \begin{bmatrix} \frac{1}{2} & 2^k \\ 0 & \frac{1}{2} \end{bmatrix}.$$

i) Show that the matrix $A(k)$ has constant eigenvalues with absolute value smaller than one. [3 marks]

ii) Determine the state transition matrix of the system, i.e. the matrix $\Phi(k,0)$ (we assume that the initial time $k_0 = 0$) such that

$$\Phi(0,0) = I, \qquad x(k) = \Phi(k,0)x(0).$$

[6 marks]

iii) Determine the set of initial conditions such that

$$\lim_{k\to\infty} x(k) = \begin{bmatrix} 0 \\ 0 \end{bmatrix}.$$

Hence, conclude that the considered linear, time-varying system, is not asymptotically stable. [5 marks]

2. Consider the discrete-time system

$$\begin{bmatrix} x_1(k+1) \\ x_2(k+1) \end{bmatrix} = \begin{bmatrix} a & 0 \\ b & c \end{bmatrix} \begin{bmatrix} x_1(k) \\ x_2(k) \end{bmatrix} + \begin{bmatrix} d \\ 0 \end{bmatrix} u(k),$$
$$y(k) = \begin{bmatrix} 0 & e \end{bmatrix} \begin{bmatrix} x_1(k) \\ x_2(k) \end{bmatrix},$$

with the initial state

$$\begin{bmatrix} x_1(0) \\ x_2(0) \end{bmatrix} = \begin{bmatrix} x_{10} \\ x_{20} \end{bmatrix},$$

where x_{10}, x_{20}, a, b, c, d and e are constant.

a) Determine for which values of

i) a, b, c and d the system is reachable. [2 marks]

ii) a, b, c and e the system is observable. [2 marks]

b) For which values of a, b, c and d the system is controllable. [8 marks]

c) Let $a = 0$ and $b = c = d = e = 1$.

i) Determine the sequence of inputs $\{u(k)\}$ to reach the state

$$\begin{bmatrix} x_1(2) \\ x_2(2) \end{bmatrix} = \begin{bmatrix} 1 \\ 1 \end{bmatrix}$$

starting from any initial condition. [2 marks]

ii) Determine the initial state $x(0)$ and the states $x(1)$ and $x(2)$, which are consistent with the values $y(0) = 1$, $y(1) = 2$, $u(0) = 2$ and $u(1) = 1$. [2 marks]

d) Let $b = 0$ and $a = d = 1$. Determine, if possible, the sequence of inputs $\{u(k)\}$ to control the state to zero for

i) $c = 0$; [2 marks]

ii) $c = 1$. [2 marks]

e) Let $a = b = c = d = 1$.

i) Design a state feedback control law $u = Kx$ placing all closed-loop eigenvalues at $\frac{1}{2}$. [4 marks]

ii) Suppose that the actuator is defective and it provides a control signal which is only a fraction of u, that is, $u = \alpha Kx$, with $\alpha \in (0, 1]$, and K as computed in part e.i). Study the stability properties of the closed-loop system as a function of α.
Hint: A polynomial $\lambda^2 + \beta\lambda + \gamma$ has all its roots with absolute value less than one if these three conditions hold concurrently

$$\begin{cases} \gamma < 1, \\ \gamma > -1 + \beta, \\ \gamma > -1 - \beta. \end{cases}$$

[6 marks]

3. Consider a linear, continuous-time, system described by the equations

$$\dot{x} = Ax + Bu \qquad y = Cx$$

with

$$A = \begin{bmatrix} -2 & 1 \\ 2 & -1 \end{bmatrix}, \qquad B = \begin{bmatrix} 1 \\ 0 \end{bmatrix}, \qquad C = \begin{bmatrix} 1 & 1 \end{bmatrix}.$$

a) Show that the system is not observable and compute the unobservable modes. Show that the system is controllable. [5 marks]

b) Design a state feedback control law $u = Kx$ such that the matrix $A + BK$ has eigenvalues equal to -2 and -1. [5 marks]

c) Design an observer such that the matrix $A + LC$ has eigenvalues equal to -2 and -3. Explain why this problem is solvable despite the fact that the system is not observable. [5 marks]

d) Using the separation principle, and the results in parts b) and c) write the equations of a dynamic, output feedback, control law which stabilizes the closed-loop system. Determine the eigenvalues of the resulting closed-loop system. [5 marks]

e) Suppose the response of the closed-loop system in part d) is *too slow*, hence it is necessary to modify the design to achieve a faster response. Suppose, in addition that the designer can either redesign the state feedback or the observer (he/she cannot redesign both).
Discuss which design has to be modified, and determine a new design achieving the fastest possible response. [5 marks]

f) Consider now the modified matrix

$$A = \begin{bmatrix} 2 & -1 \\ -2 & 1 \end{bmatrix}$$

and the same matrices B and C as already defined. Compute the input-output transfer function $\dfrac{Y(s)}{U(s)}$ and the input-state transfer vector $\dfrac{X(s)}{U(s)}$. Based on these two transfer functions, explain what the expected behaviour of $y(t)$ and $x(t)$ is when the input $u(t)$ is a Dirac impulse (meaning that $U(s) = 1$) and comment on whether the system is stable, asymptotically stable or unstable. [5 marks]